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EXIT DEVICE WITH SCISSOR BLOCKING

Abstract

An exemplary exit device generally includes a pushbar, a scissor mechanism, a blocking member, and a driver. The pushbar is mounted for movement between a projected position and a depressed position. The scissor mechanism is connected between the pushbar and a latch mechanism, and depression of the pushbar causes an inward scissoring movement of the scissor mechanism and actuation of the latch mechanism. The blocking member is mounted for movement between a blocking position, in which the blocking member prevents the inward scissoring movement, and an unblocking position, in which the blocking member does not prevent the inward scissoring movement. The driver is operable to move the blocking member between the blocking position and the unblocking position.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to exit devices, and more particularly but not exclusively relates to delayed egress exit devices.

BACKGROUND

[0002] Delayed egress systems are often installed to doors between a more-secure area and a less-secure area, and typically selectively permit egress from the more-secure area to the less-secure area. In contrast to standard free egress exit device systems, in which depression of the pushbar results in immediate actuation of the latch mechanism, delayed egress systems typically delay the actuation of the latch mechanism for a predetermined time period. Example facilities that may utilize such delayed egress systems include behavioral health centers, memory care centers, maternity wards, retail environments, and airports. Many conventional delayed egress systems utilize maglocks, which requires a separate unit to be installed to the door, increasing wiring and installation complexities. For these reasons among others, there remains a need for further improvements in this technological field.

SUMMARY

[0003] An exemplary exit device generally includes a pushbar, a scissor mechanism, a blocking member, and a driver. The pushbar is mounted for movement between a projected position and a depressed position. The scissor mechanism is connected between the pushbar and a latch mechanism, and depression of the pushbar causes an inward scissoring movement of the scissor mechanism and actuation of the latch mechanism. The blocking member is mounted for movement between a blocking position, in which the blocking member prevents the inward scissoring movement, and an unblocking position, in which the blocking member does not prevent the inward scissoring movement. The driver is operable to move the blocking member between the blocking position and the unblocking position. Further embodiments, forms, features, and aspects of the present application shall become apparent from the description and figures provided herewith.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0004] FIG. 1 is a perspective illustration of an exit device according to certain embodiments installed to a door.

[0005] FIG. 2 is a perspective view of a portion of the exit device.

[0006] FIG. 3 is a cross-sectional illustration of a portion of the exit device, including a proximal scissor mechanism and a portion of a blocking mechanism.

[0007] FIG. 4 is a perspective illustration of a portion of the exit device, including the proximal scissor mechanism and a portion of the blocking mechanism.

[0008] FIG. 5 is a cross-sectional illustration of a portion of the exit device, including a distal scissor mechanism and a request-to-exit sensor.

[0009] FIG. 6 is a perspective illustration of a portion of the exit device, including the distal scissor mechanism and the request-to-exit sensor.

[0010] FIG. 7 is a schematic block diagram of the exit device.

[0011] FIG. 8 is a cross-sectional illustration of a portion of the exit device with the exit device in a locked state.

[0012] FIG. 9 is a cross-sectional illustration of a portion of the exit device with the exit device in an unlocked state.

[0013] FIG. 10 is a cross-sectional illustration of a portion of the exit device with the exit device in an actuated state.

[0014] FIG. 11 is a schematic flow diagram of a process according to certain embodiments.

[0015] FIG. 12 is a schematic block diagram of a computing device according to certain embodiments.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0016] Although the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

[0017] References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. It should further be appreciated that although reference to a “preferred” component or feature may indicate the desirability of a particular component or feature with respect to an embodiment, the disclosure is not so limiting with respect to other embodiments, which may omit such a component or feature. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0018] Additionally, it should be appreciated that items included in a list in the form of “at least one of A, B, and C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Items listed in the form of “A, B, and/or C” can also mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Further, with respect to the claims, the use of words and phrases such as “a,” “an,” “at least one,” and/or “at least one portion” should not be interpreted so as to be limiting to only one such element unless specifically stated to the contrary, and the use of phrases such as “at least a portion” and/or “a portion” should be interpreted as encompassing both embodiments including only a portion of such element and embodiments including the entirety of such element unless specifically stated to the contrary.

[0019] In the drawings, some structural or method features may be shown in certain specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not necessarily be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures unless indicated to the contrary. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may be omitted or may be combined with other features.

[0020] The disclosed embodiments may, in some cases, be implemented in hardware, firmware, software, or a combination thereof. The disclosed embodiments may also be implemented as instructions carried by or stored on one or more transitory or non-transitory machine-readable (e.g., computer-readable) storage media, which may be read and executed by one or more processors. A machine-readable storage medium may be embodied as any storage device, mechanism, or other physical structure for storing or transmitting information in a form readable by a machine (e.g., a volatile or non-volatile memory, a media disc, or other media device).

[0021] With reference to FIG. 1, illustrated therein is a door **90** having installed thereto an exit device **100** according to certain embodiments, which in the illustrated form is provided as a delayed egress exit device. The door **90** has an egress side **91**, which typically faces a more-secure area, and a non-egress side **92**, which typically faces a less-secure area. As described herein, the delayed egress exit device **100** is operable to provide for delayed egress through the door **90** in response to a user-initiated exit request, and may further be configured to provide an alert during the delay

period. While certain concepts are described herein with reference to a delayed egress embodiment of the exit device **100**, it should be appreciated that the exit device **100** may be provided in another form.

[0022] With additional reference to FIG. **2**, the illustrated exit device **100** generally includes a mounting assembly **110**, a latch mechanism **120**, and a drive assembly **130** operable to selectively retract the latch mechanism **120**. As described herein, the drive assembly **130** includes, among other elements and features, a proximal scissor mechanism **140** and a distal scissor mechanism **150**. The exit device **100** further includes a blocking mechanism **160** operable to selectively prevent the drive assembly **130** from retracting the latch mechanism **120**, a driver **170** operable to move the blocking mechanism **160** between a blocking state and an unblocking state, and a control assembly **180** configured to control operation of the driver **170**.

[0023] The mounting assembly **110** is configured for mounting to the door **90**, and generally includes a channel member **111**, a base plate **112** mounted in the channel member **111**, a header plate **113** positioned at a proximal end of the base plate **112**, a header case **114** covering the header plate **113**, and a mounting bracket **115** mounted to the header plate **113** within the header case **114**. In the illustrated form, the mounting assembly **110** further includes a spring anchor **116**, which is secured to the base plate **112** and provides an anchor point for a spring **106** as described herein.

[0024] With additional reference to FIG. **3**, the latch mechanism **120** includes a latchbolt **122** having an extended position and a retracted position. In the illustrated form, the latchbolt **122** is pivotably mounted to the mounting bracket **115**, is connected with the drive assembly **130** via a retractor **124**, and is biased toward its extended position, for example by a spring **126**. As described herein, actuation of the drive assembly **130** causes a corresponding actuation of the latch mechanism **120** for retraction of the latchbolt **122**.

[0025] In the illustrated form, the exit device **100** is provided as a rim format exit device, in which the latch mechanism **120** is mounted within the header case **114**. It is also contemplated that the concepts described herein may be utilized in combination with other forms of exit devices. As one example, the exit device **100** may be provided in the form of a vertical exit device, in which at least one latch mechanism is positioned remotely from (e.g., above or below) the header case **114**.

Additionally or alternatively, a latch mechanism may be provided in a mortise case configured for installation within the door **90**. In such forms, the latch mechanism may nonetheless be selectively actuated by the drive assembly **130** in a manner analogous to that described herein.

[0026] The drive assembly **130** generally includes a pushbar **132** mounted for movement between an unactuated or projected position and an actuated or depressed position. As described herein, actuation of the drive assembly **130** by depressing the pushbar **132** causes a corresponding actuation of the latch mechanism **120** for retraction of the latchbolt **122**. As noted above, the drive assembly **130** includes a pair of scissor mechanisms **140**, **150**. These scissor mechanisms **140**, **150** support the pushbar **132** for movement between the projected position and the depressed position. The scissor mechanisms include a proximal scissor mechanism **140** that is positioned nearer the header case **114**, and a distal scissor mechanism **150** that is positioned farther from the header case **114**. The drive assembly **130** also includes a link plate **134** operable to retract the retractor **124**, a latch link **136** operable to retract the link plate **134**, and a scissor connector **138** coupled with the latch link **136**. As described herein, the latch link **136** is engaged between the proximal scissor mechanism **140** and the link plate **134** such that depression of the pushbar **132** retracts the latch link **136** and the link plate **134**, thereby actuating the latch mechanism **120** and retracting the latchbolt **122**.

[0027] With additional reference to FIG. **4**, the proximal scissor mechanism **140** generally includes a support arm **141** pivotably mounted to the mounting assembly **110**, and a drive arm **144** pivotably connected between the pushbar **132** and the latch link **136**. The support arm **141** includes a first end portion **142** pivotably connected to the mounting assembly **110**, and an opposite second end portion **143** that is pivotably coupled with the drive arm **144**. The drive arm **144** includes a first end portion

145 pivotably connected with the pushbar **132**, and an opposite second end portion **146** that is pivotably connected with both the latch link **136** and the scissor connector **138**. The support arm **141** meets the drive arm **144** at a scissor angle **0140** that varies during movement of the drive assembly **130**. Each of the support arm **141** and the drive arm **144** includes a corresponding lobe **147**, and the lobes **147** face generally toward one another such that a gap **148** is formed therebetween.

[0028] During unblocked movement of the scissor mechanism **140**, depression of the pushbar **132** causes the drive arm first end portion **145** to move toward the base plate **112** (downward in FIG. 3), while distally driving the drive arm second end portion **146** away from the header case **114** for retraction of the latch link **136**. This action also pivots the support arm **141**, thereby causing the arms **141**, **144** to scissor in a closing direction. Scissoring in the closing direction reduces the scissor angle θ **140**, and closes the gap **148** as the latch link **136** actuates the latch mechanism **120**. As described herein, the blocking mechanism **160** is configured to selectively prevent the inward scissoring action in order to prevent depression of the pushbar **132** and retraction of the latchbolt **122**.

[0029] With additional reference to FIGS. 5 and 6, the distal scissor mechanism **150** is similar to the proximal scissor mechanism **140**, and similar reference characters are used to indicate similar elements and features. For example, the distal scissor mechanism **150** includes a support arm **151** pivotably mounted to the mounting assembly **110**, and a drive arm **154** pivotably connected between the pushbar **132** and the scissor connector **138**. The proximal end of the scissor connector **138** is coupled with the latch link **136** such that depression of the pushbar **132** causes the distal scissor mechanism **150** to aid in retraction of the latch link **136** by retracting the transmission link **158**.

[0030] Mounted to the distal scissor mechanism **150** is an actuating arm **108** operable to engage a request-to-exit (REX) sensor **184** of the control assembly **180**, which in the illustrated form is mounted to the underside of the pushbar **132** for movement with the pushbar **132**. In the illustrated embodiment, the REX sensor **184** is provided in the form of a mechanical snap action switch that is actuated in response to depression of a spring arm **185**. It is also contemplated that the REX sensor **184** may take another form, such as one involving a reed switch, a Hall effect sensor, an optical sensor, or another form of sensor. As described herein, the blocking mechanism **160** is operable to selectively prevent the pushbar **132** from being depressed by an amount sufficient to actuate the latch mechanism **120**. However, even when the blocking mechanism **160** is in its blocking position, the pushbar **132** remains capable of slight movement. When an actuating force is applied by a user to the pushbar **132**, this slight movement causes the REX sensor **184** to move downward by an amount sufficient to cause the REX sensor **184** to transmit a REX signal (e.g., by causing the actuating arm **108** to depress the spring arm **185**). As described herein, the control assembly **180** may actuate the driver **170** to move the blocking mechanism **160** based at least in part upon receipt of the REX signal.

[0031] The blocking mechanism **160** generally includes a blocking bracket **162** having a blocking member **164** mounted thereto, and a driver link **166** connected between the blocking bracket **162** and the driver **170**. As described herein, the blocking mechanism **160** has a blocking position (FIG. 8) and an unblocking position (FIG. 9), and may be biased toward the unblocking position. While other forms of biasing are contemplated, in the illustrated form, a blocking mechanism spring **106** is engaged between the spring anchor **116** and the driver link **166** such that the spring **106** biases the driver link **166** in the proximal direction to thereby bias the blocking mechanism **160** toward its unblocking position.

[0032] The blocking bracket **162** is coupled with the proximal end of the driver link **166**, and may include a guide slot **163** that engages a guide pin **103** to aid in guiding movement of the blocking bracket **162** between its blocking position and its unblocking position. In the illustrated form, the blocking member **164** comprises a pin **164'** that is operable to be received in the gap **148** to thereby

block the scissoring action of the proximal scissor mechanism **140**. In certain embodiments, the blocking member **164** may include a bushing **165** that discourages direct contact between the pin **164'** and the scissor mechanism **140**. In certain forms, the bushing **165** may be formed of a non-metallic (e.g., plastic or elastomeric) material to reduce sounds and/or vibrations resulting from contact between the blocking member **164** and the scissor mechanism **140**. In certain forms, the bushing **165** may be rotatable relative to the bracket **162** such that the blocking member **164** is provided in the form of a roller **168**.

[0033] The driver **170** is operable to move the blocking mechanism **160** between its blocking position and its unblocking position, and generally includes a body **172** and a reciprocating shaft **174** that is coupled with a distal end portion **167** of the driver link **166**. In the illustrated form, the driver **170** is provided in the form of a linear motor. It is also contemplated that the driver **170** may be provided in another form, such as one including a solenoid or electromagnet.

[0034] As noted above the illustrated blocking mechanism **160** is biased toward its unblocking position, in which the blocking mechanism **160** does not prevent depression of the pushbar **132** and the resulting actuation of the latch mechanism **120**. During normal use of the exit device **100**, however, it is typically desirable to maintain the blocking mechanism **160** in its blocking position to prevent such free egress. Accordingly, the driver **170** may be normally powered during operation of the exit device **100** to thereby maintain the blocking mechanism **160** in the blocking position against the force of the spring **106**. As described herein, the control assembly **180** may selectively cut power to the driver **170** to permit the spring **106** to drive the blocking mechanism **160** toward its unblocking position to permit egress.

[0035] With additional reference to FIG. 7, the control assembly **180** is in communication with the driver **170**, and generally includes control circuitry **182**, the REX sensor **184**, and a delay timer **186**, and may further include an alert mechanism **188**. The control assembly **180** may be configured for connection to line power **82**, and is configured to normally power the driver **170** to thereby selectively maintain the blocking mechanism **160** in its blocking position. When a user depresses the pushbar **132** and actuates the REX sensor **184**, the control circuitry **182** initiates the delay timer **186**. Upon expiration of the delay timer **186**, the control circuitry **182** cuts power to the driver **170** to thereby cause the blocking mechanism **160** to move to its blocking position under the biasing force of the spring **106**.

[0036] When the user depresses the pushbar **132**, the control circuitry **182** may activate the alert mechanism **188**. In certain embodiments, the alert mechanism **188** may be configured to provide a local alert, such as an audible and/or visual alert. For example, the alert mechanism **188** may include a buzzer and/or a flashing light, either of which may alert users in the area of the exit request. It is also contemplated that the alert mechanism **188** may be configured to provide a remote alert, for example by sending an alert signal to an access control system and/or a security station. In certain embodiments, activating the alert mechanism **188** may result in storing the time of the exit request in an audit trail.

[0037] In certain embodiments, the control circuitry **182** may comprise a controller **183**, which may be embodied as a computing device as described herein with reference to FIG. 11. However, it should be appreciated that, in at least certain embodiments, the control circuitry **182** may not necessarily include a controller or other computing device, and may instead include traditional non-integrated circuits. Moreover, it should be appreciated that in certain embodiments, the control circuitry **182** may be considered to include the REX sensor **184**, the delay timer **186**, and/or the alert mechanism **188**.

[0038] With additional reference to FIGS. 8-10, the illustrated exit device **100** has a delayed egress mode in which the exit device **100** is operable in a plurality of states, including a locked state (FIG. 8), an unlocked state (FIG. 9), and an actuated state (FIG. 10). As described herein, when operating in the delayed egress mode, the exit device **100** normally occupies the locked state, and selectively transitions to the unlocked state to permit actuation of the latch mechanism **120** by the drive

assembly **130** for egress through the door **90**.

[0039] With the exit device **100** operating in the delayed egress mode, the control assembly **180** by default maintains the blocking mechanism **160** in the blocking position to thereby retain the exit device **100** in its locked state (FIG. **8**). In the illustrated form, maintaining the exit device **100** in its locked state involves powering the driver **170** to overcome the bias force of the spring **106** and maintain the blocking mechanism **160** in its blocking position. With the blocking mechanism **160** in its blocking position, the blocking member **164** is received in the gap **148** and captured between the lobes **147** to thereby prevent inward scissoring of the proximal scissor mechanism **140**. As a result, the blocking mechanism **160** prevents the full depression of the pushbar **132** that would otherwise actuate the latch mechanism **120**.

[0040] When a user desires to exit the secured area via the door **90**, the user may exert a depressing force on the pushbar **132** to thereby provide the exit device **100** with an exit request that is registered by the REX sensor **184**. In the illustrated form, the depressing force results in a slight movement of the pushbar **132** that activates the REX sensor **184** as described above. It is also contemplated that the user may register an exit request in another manner, such as by activating a remote switch, presenting a credential, or providing another indication of an intent and/or request to exit.

[0041] Upon receiving the REX signal from the REX sensor **184**, the control circuitry **182** may initiate the delay timer **186**. The delay timer **186** may, for example, be set to a time period between three seconds and ten seconds, such as a time period of about five seconds. During the delay period, the control circuitry **182** maintains the exit device **100** in the locked state, for example by maintaining power to the driver **170** such that the driver **170** continues to hold the blocking mechanism **160** in its blocking state. During the delay period, the control circuitry **182** may activate the alert mechanism **188** to provide an alert (e.g., an audible alert, a visual alert, and/or an electronic alert) regarding the exit request.

[0042] Upon expiration of the delay timer **186**, the control circuitry **182** causes the exit device **100** to transition to the unlocked state (FIG. **9**). For example, the control circuitry **182** may cut power to the driver **170** such that the spring **106** drives the blocking mechanism **160** toward its unblocking position, thereby removing the blocking member **164** from the gap **148** between the lobes **147**. With the blocking mechanism **160** in its unblocking position, the blocking member **164** is removed from the gap **148** and no longer prevents the inward scissoring motion of the scissor mechanism **140**. As a result, the user is able to depress the pushbar **132**.

[0043] As the user depresses the pushbar **132**, the exit device **100** transitions to the actuated state (FIG. **10**). More particularly, the inward scissoring of the scissor mechanisms **140**, **150** drives the scissor connector **138** and the latch link **136** in the distal direction, thereby actuating the latch mechanism **120** as described above. With the latch mechanism **120** actuated, the door **90** can be opened to permit egress from the secured area.

[0044] After causing the blocking mechanism **160** to move to the unblocking position, the control circuitry **182** may cause the blocking mechanism **160** to return to the blocking position in response to a return criterion, such as expiration of a second delay timer. For example, the control circuitry **182** may once again power the driver **170** to cause the driver **170** to overcome the bias force of the spring **106** to return the blocking mechanism **160** to its blocking position, thereby returning the exit device **100** to its locked state (FIG. **8**).

[0045] With additional reference to FIG. **11**, an exemplary process **200** that may be performed using the exit device **100** is illustrated. Blocks illustrated for the processes in the present application are understood to be examples only, and blocks may be combined or divided, and added or removed, as well as re-ordered in whole or in part, unless explicitly stated to the contrary. Unless specified to the contrary, it is contemplated that certain blocks performed in the process **200** may be performed wholly by the control circuitry **182**, a controller **183**, the REX sensor **184**, the delay timer **186**, the alert mechanism **188**, and/or the driver **170**, or that the blocks may be distributed

among one or more of the elements and/or additional devices or systems that are not specifically illustrated in FIGS. **1-10**. Additionally, while the blocks are illustrated in a relatively serial fashion, it is to be understood that two or more of the blocks may be performed concurrently or in parallel with one another. Moreover, while the process **200** is described herein with specific reference to the exit device **100** illustrated in FIGS. **1-10**, it should be appreciated that the process **200** may be performed with exit devices having additional and/or alternative features.

[0046] The process **200** generally involves delaying egress via a door having an exit device mounted thereon, the exit device including a blocking member, a scissor mechanism, and a pushbar. For example, the process **200** may involve delaying egress via the door **90**, on which is mounted an exit device **100** that includes a blocking member **164**, a scissor mechanism **150**, and a pushbar **132**.

[0047] The process **200** may include block **210**, which generally involves selectively maintaining the exit device in a locked state. In certain embodiments, block **210** may include block **212**, which generally involves maintaining the blocking member in a blocking position in which the blocking member prevents an inward scissoring movement of the scissor mechanism, thereby preventing movement of the pushbar from a projected position to a depressed position. For example, block **212** may involve maintaining the blocking member **164** in the blocking position (FIG. **8**), in which the blocking member **164** prevents inward scissoring movement of the scissor mechanism **140**. As described above, this prevention of the inward scissoring prevents movement of the pushbar **132** from its projected position (FIGS. **8** and **9**) to its depressed position (FIG. **10**). While other forms are contemplated, in the illustrated form, the blocking member **164** includes a pin **164'** and a bushing **165**, and the process **200** further includes the bushing **165** preventing direct contact between the scissor mechanism **140** and the pin **164'**.

[0048] As noted above, in certain embodiments, the blocking member **164** may be biased toward its unblocking position, for example by a spring **106**. In such forms, block **210** may include block **214**, which generally involves powering a driver connected with the blocking member such that the driver overcomes the biasing and retains the blocking member in the blocking position. For example, block **210** may involve powering the driver **170** as described above such that the driver **170** overcomes the biasing of the spring **106** and retains the blocking member **164** in the blocking position (FIG. **8**) against the biasing force of the spring **106**.

[0049] Block **210** may be performed to maintain the exit device **100** in the locked state (FIG. **8**) until an exit request **202** is received by the exit device **100**. The exit request **202** may, for example, be generated by the REX sensor **184** in response to a user attempting to depress the pushbar **132** as described above. In response to the exit request **202**, the process **200** may continue to block **220**.

[0050] Block **220** may be performed in response to the exit request **202**, and generally involves continuing to maintain the exit device in the locked state for a predetermined delay period. Block **220** may include block **222**, which generally involves maintaining the blocking member **164** in the blocking position, for example by powering the driver **170** in block **224**. In certain forms, the process **200** may involve performing block **220** for the duration of a delay period **203**, which may begin by activating the delay timer **186** in response to the exit request **202**. In certain forms, block **220** may include block **226**, which generally involves activating an alert mechanism **188** to thereby provide a local and/or remote alert regarding the exit request **202**.

[0051] In response to expiration of the delay period **203**, the process **200** may continue to block **230**, which generally involves moving the exit device to an unlocked state. In certain forms, block **230** may involve block **232**, which generally involves moving the blocking member to the unblocking position, for example by cutting power to the driver in block **234**. By way of illustration, blocks **232** and **234** may involve moving the blocking member **164** to its unblocking position (FIG. **9**) in response to the control assembly **180** cutting power to the driver **170**. As described above, moving the blocking member **164** to its unblocking position (FIG. **9**) causes the blocking member **164** to permit the inward scissoring movement of the scissor mechanism **140**,

thereby permitting depression of the pushbar **132**.

[0052] In certain forms, the process **200** may involve guiding movement of the blocking member **164** between its blocking position and its unblocking position. In the illustrated form, the blocking member **164** is mounted to a movable bracket **162** including a guide slot **163**, and the process **200** involves guiding movement of the blocking member **164** via the guide slot **163** and a guide pin **103**, which is received in the guide slot **163**.

[0053] With the exit device **100** in its unlocked state, a user may depress the pushbar **132**. In response to such pushbar depression **204**, the process **200** may continue to block **240**, which generally involves actuating a latch mechanism to thereby permit egress via the door. In certain embodiments, the latch mechanism actuated in block **240** may be a latch mechanism of a rim format exit device, such as the latch mechanism **120** illustrated in connection with the illustrative exit device **100**. It is also contemplated that the latch mechanism actuated in block **240** may be a remote latch mechanism, such as a remote latch mechanism of a vertical format version of the exit device **100**.

[0054] Following the unlocking of the exit device in block **230**, a relock condition **205** may occur. In certain embodiments, the relock condition **205** may involve a relock timer. For example, the control assembly **180** may initiate a relock timer during block **230**, and expiration of the relock timer may be interpreted as the relock condition **205**. In response to the relock condition **205**, the process **200** may return to block **210**, in which the exit device **100** is once again maintained in the locked state in anticipation of the next exit request **202**.

[0055] Referring now to FIG. **11**, a simplified block diagram of at least one embodiment of a computing device **300** is shown. The illustrative computing device **300** depicts at least one embodiment of a controller that may be utilized in connection with the controller **183** illustrated in FIG. **7**.

[0056] Depending on the particular embodiment, the computing device **300** may be embodied as a server, desktop computer, laptop computer, tablet computer, notebook, netbook, Ultrabook™, mobile computing device, cellular phone, smartphone, wearable computing device, personal digital assistant, Internet of Things (IoT) device, reader device, access control device, control panel, processing system, router, gateway, and/or any other computing, processing, and/or communication device capable of performing the functions described herein.

[0057] The computing device **300** includes a processing device **302** that executes algorithms and/or processes data in accordance with operating logic **308**, an input/output device **304** that enables communication between the computing device **300** and one or more external devices **310**, and memory **306** which stores, for example, data received from the external device **310** via the input/output device **304**.

[0058] The input/output device **304** allows the computing device **300** to communicate with the external device **310**. For example, the input/output device **304** may include a transceiver, a network adapter, a network card, an interface, one or more communication ports (e.g., a USB port, serial port, parallel port, an analog port, a digital port, VGA, DVI, HDMI, Fire Wire, CAT 5, or any other type of communication port or interface), and/or other communication circuitry. Communication circuitry may be configured to use any one or more communication technologies (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Bluetooth Low Energy (BLE), Wi-Fi®, WiMAX, etc.) to effect such communication depending on the particular computing device **300**. The input/output device **304** may include hardware, software, and/or firmware suitable for performing the techniques described herein.

[0059] The external device **310** may be any type of device that allows data to be inputted or outputted from the computing device **300**. For example, in various embodiments, the external device **310** may be embodied as the driver **170**, the REX sensor **184**, the delay timer **186**, and/or the alert mechanism **188**. Further, in some embodiments, the external device **310** may be embodied as another computing device, switch, diagnostic tool, controller, printer, display, alarm, peripheral

device (e.g., keyboard, mouse, touch screen display, etc.), and/or any other computing, processing, and/or communication device capable of performing the functions described herein. Furthermore, in some embodiments, it should be appreciated that the external device **310** may be integrated into the computing device **300**.

[0060] The processing device **302** may be embodied as any type of processor(s) capable of performing the functions described herein. In particular, the processing device **302** may be embodied as one or more single or multi-core processors, microcontrollers, or other processor or processing/controlling circuits. For example, in some embodiments, the processing device **302** may include or be embodied as an arithmetic logic unit (ALU), central processing unit (CPU), digital signal processor (DSP), and/or another suitable processor(s). The processing device **302** may be a programmable type, a dedicated hardwired state machine, or a combination thereof. Processing devices **302** with multiple processing units may utilize distributed, pipelined, and/or parallel processing in various embodiments. Further, the processing device **302** may be dedicated to performance of just the operations described herein, or may be utilized in one or more additional applications. In the illustrative embodiment, the processing device **302** is of a programmable variety that executes algorithms and/or processes data in accordance with operating logic **308** as defined by programming instructions (such as software or firmware) stored in memory **306**. Additionally or alternatively, the operating logic **308** for processing device **302** may be at least partially defined by hardwired logic or other hardware. Further, the processing device **302** may include one or more components of any type suitable to process the signals received from input/output device **304** or from other components or devices and to provide desired output signals. Such components may include digital circuitry, analog circuitry, or a combination thereof.

[0061] The memory **306** may be of one or more types of non-transitory computer-readable media, such as a solid-state memory, electromagnetic memory, optical memory, or a combination thereof. Furthermore, the memory **306** may be volatile and/or nonvolatile and, in some embodiments, some or all of the memory **306** may be of a portable variety, such as a disk, tape, memory stick, cartridge, and/or other suitable portable memory. In operation, the memory **306** may store various data and software used during operation of the computing device **300** such as operating systems, applications, programs, libraries, and drivers. It should be appreciated that the memory **306** may store data that is manipulated by the operating logic **308** of processing device **302**, such as, for example, data representative of signals received from and/or sent to the input/output device **304** in addition to or in lieu of storing programming instructions defining operating logic **308**. As illustrated, the memory **306** may be included with the processing device **302** and/or coupled to the processing device **302** depending on the particular embodiment. For example, in some embodiments, the processing device **302**, the memory **306**, and/or other components of the computing device **300** may form a portion of a system-on-a-chip (SoC) and be incorporated on a single integrated circuit chip.

[0062] In some embodiments, various components of the computing device **300** (e.g., the processing device **302** and the memory **306**) may be communicatively coupled via an input/output subsystem, which may be embodied as circuitry and/or components to facilitate input/output operations with the processing device **302**, the memory **306**, and other components of the computing device **300**. For example, the input/output subsystem may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations.

[0063] The computing device **300** may include other or additional components, such as those commonly found in a typical computing device (e.g., various input/output devices and/or other components), in other embodiments. It should be further appreciated that one or more of the components of the computing device **300** described herein may be distributed across multiple computing devices. In other words, the techniques described herein may be employed by a

computing system that includes one or more computing devices. Additionally, although only a single processing device **302**, I/O device **304**, and memory **306** are illustratively shown in FIG. **11**, it should be appreciated that a particular computing device **300** may include multiple processing devices **302**, I/O devices **304**, and/or memories **306** in other embodiments. Further, in some embodiments, more than one external device **310** may be in communication with the computing device **300**.

[0064] While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only the preferred embodiments have been shown and described and that all changes and modifications that come within the spirit of the inventions are desired to be protected.

[0065] It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicate that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention, the scope being defined by the claims that follow. In reading the claims, it is intended that when words such as “a,” “an,” “at least one,” or “at least one portion” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. When the language “at least a portion” and/or “a portion” is used the item can include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. An exit device, comprising: a mounting assembly configured for mounting to a door; a scissor mechanism movably mounted to the mounting assembly, the scissor mechanism defining a gap that expands during an outward scissoring movement and contracts during an inward scissoring movement, wherein the scissor mechanism is operably connected with a latch mechanism such that the inward scissoring movement causes a corresponding actuation of the latch mechanism; a pushbar operably connected to the scissor mechanism such that depression of the pushbar causes the inward scissoring movement; a blocking mechanism operable to selectively prevent the inward scissoring movement, wherein the blocking mechanism has a blocking position in which a blocking member is positioned in the gap and prevents the inward scissoring movement, and wherein the blocking mechanism has an unblocking position in which the blocking member does not prevent the inward scissoring movement; and a driver operable to move the blocking mechanism between the blocking position and the unblocking position.
2. The exit device of claim 1, wherein the blocking member comprises a pin and a bushing mounted to the pin.
3. The exit device of claim 2, wherein the bushing is formed of a non-metallic material.
4. The exit device of claim 2, wherein the bushing is rotatable.
5. The exit device of claim 1, further comprising a control assembly configured to cause the driver to selectively maintain the blocking mechanism in the blocking position, and to cause the driver to move the blocking mechanism to the unblocking position a predetermined amount of time following an exit request.
6. The exit device of claim 5, wherein the control assembly comprises a request-to-exit sensor configured to generate the exit request when actuated by a user.
7. The exit device of claim 1, wherein the scissor mechanism includes a pair of lobes, and wherein the gap is defined between the pair of lobes.
8. The exit device of claim 1, wherein the blocking mechanism further comprises a bracket connected between the driver and the blocking member; wherein the bracket comprises a guide slot that receives a guide pin to thereby guide movement of the blocking mechanism.
9. The exit device of claim 1, wherein the blocking mechanism is biased toward the unblocking

state.

10. A method of delaying egress via a door having an exit device mounted thereon, the exit device comprising a blocking member, a scissor mechanism, and a pushbar, the method comprising: selectively maintaining the exit device in a locked state, comprising maintaining the blocking member in a blocking position in which the blocking member prevents an inward scissoring movement of the scissor mechanism, thereby preventing movement of the pushbar from a projected position to a depressed position; in response to receiving an exit request, continuing to maintain the exit device in the locked state for a predetermined delay period; in response to expiration of the delay period, transitioning the exit device to an unlocked state, comprising moving the blocking member to an unblocking position in which the blocking member permits the inward scissoring movement, thereby permitting depression of the pushbar; and in response to depression of the pushbar, actuating a latch mechanism to thereby permit egress via the door.

11. The method of claim 10, further comprising biasing the blocking member toward the unblocking position.

12. The method of claim 11, wherein selectively maintaining the exit device in a locked state further comprises powering a driver connected with the blocking member such that the driver overcomes the biasing and retains the blocking member in the blocking position.

13. The method of claim 10, wherein the blocking member comprises a pin and a bushing mounted to the pin; and wherein the method further comprises the bushing preventing direct contact between the scissor mechanism and the pin.

14. The method of claim 10, wherein the blocking member is mounted to a movable bracket; and wherein the method further comprises guiding movement of the blocking member via a guide slot formed in the movable bracket and a guide pin received in the guide slot.

15. An exit device, comprising: a pushbar mounted for movement between a projected position and a depressed position; a scissor mechanism connected between the pushbar and a latch mechanism, wherein depression of the pushbar causes an inward scissoring movement of the scissor mechanism and actuation of the latch mechanism; a blocking member mounted for movement between a blocking position, in which the blocking member prevents the inward scissoring movement, and an unblocking position, in which the blocking member does not prevent the inward scissoring movement; and a driver operable to move the blocking member between the blocking position and the unblocking position.

16. The exit device of claim 15, further comprising a control assembly configured to cause the driver to selectively maintain the blocking mechanism in the blocking position, and to cause the driver to move the blocking mechanism to the unblocking position a predetermined amount of time following an exit request.

17. The exit device of claim 16, wherein the control assembly comprises a request-to-exit sensor configured to generate the exit request when actuated by a user.

18. The exit device of claim 15, wherein the scissor mechanism comprises a first arm and a second arm; wherein each of the first arm and the second arm comprises a corresponding and respective lobe; wherein a gap is formed between the lobes when the scissor mechanism is in an unactuated condition; wherein the blocking member is received in the gap when in the blocking position; and wherein the blocking member is removed from the gap when in the unblocking position.

19. The exit device of claim 15, wherein the blocking mechanism comprises a roller.

20. The exit device of claim 15, wherein the scissor mechanism comprises a first arm and a second arm; and wherein the blocking mechanism in the blocking position is interposed between the first arm and the second arm to thereby block the inward scissoring movement of the scissor mechanism.
